LABORATORY RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	2
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1. TITLE

Design and Implementation of Webpage Styling Using CSS3 and Interactive Graphics with HTML5 Canvas

2. AIM

To design a modern, responsive web page utilizing CSS3 linear gradients, flexbox cards, box-shadows, hover animations, and the HTML5 Canvas 2D API for rendering custom shapes and text.

3. OBJECTIVE

The objectives of this experiment are:

- To apply CSS3 styling properties including background gradients and typography.
- To construct a responsive card layout using CSS3 Flexbox container properties.
- To implement interactive hover effects and box-shadow styling on HTML elements.
- To use the HTML5 Canvas 2D rendering context (getContext('2d')) for drawing primitives.
- To render geometric shapes (rectangles, arcs/circles, lines) and custom text on canvas.

4. THEORY

CSS3 introduces powerful visual capabilities such as linear gradients (background: linear-gradient(...)), flexible box layout (display: flex), drop shadows (box-shadow), and smooth transition effects on hover. Flexbox simplifies alignment and distribution of elements within a container.

HTML5 Canvas provides an immediate-mode 2D drawing surface accessed via JavaScript using document.getElementById().getContext('2d'). It allows precise programmatic drawing of shapes using methods like fillRect(), arc(), lineTo(), fill(), stroke(), and fillText().

Application Flow:

HTML5 Structure -> CSS3 Flexbox & Gradient Styling -> Canvas Element -> JavaScript 2D Context API -> Geometric Primitive Rendering -> Interactive Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document structure with header, nav bar, canvas element, and card containers.
2. Apply CSS3 reset rules and define body background using linear-gradient(white, #e8e8e8).
3. Style navigation hyperlinks with custom padding, borders, border-radius, and hover transition state.
4. Configure the <canvas> element with ID 'myCanvas', dimensions (width=450, height=200), and border styling.
5. Write JavaScript to retrieve 2D rendering context (var ctx = canvas.getContext('2d')).
6. Draw a filled blue rectangle (fillRect) with labeled text 'Rectangle' (fillText).
7. Draw a filled red circle using path creation (arc, fill) with labeled text 'Circle'.
8. Draw a green line segment (moveTo, lineTo, stroke) with labeled text 'Line'.
9. Render bold text 'Harsha Vardhan' at the bottom of the canvas.
10. Create a flexible container (.container) with flex layout and style cards (.card) with borders, rounded corners, box shadows, and button hover states.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>My Styled Webpage</title>

    <style>
        * {
            margin: 0;
            padding: 0;
        }

        body {
            font-family: Arial, Helvetica, sans-serif;
            background: linear-gradient(white, #e8e8e8);
            padding: 15px;
        }

        header {
            text-align: center;
            padding: 10px;
            font-family: Georgia, serif;
        }

        header h1 {
            font-size: 32px;
        }

        header p {
            font-size: 16px;
            font-family: Arial, sans-serif;
        }

        nav {
            text-align: center;
            padding: 10px;
        }

        nav a {
            color: black;
            text-decoration: none;
            padding: 6px 16px;
            margin: 0 5px;
            border: 1px solid #ccc;
            border-radius: 5px;
        }

        nav a:hover {
            background: #ddd;
            color: #333;
            border-color: #999;
        }

        canvas {
            border: 2px solid #333;
            border-radius: 8px;
            display: block;
            margin: 10px auto;
        }

        h2 {
            text-align: center;
            font-family: Georgia, serif;
        }

        .container {
            display: flex;
            padding: 10px 0;
        }

        .card {
            border: 2px solid #666;
```

```
            border-radius: 10px;
            padding: 15px;
            width: 180px;
            margin: 5px;
            box-shadow: 3px 3px 8px #aaa;
        }

        .card h3 {
            font-size: 20px;
            font-family: Georgia, serif;
        }

        .card p {
            font-size: 14px;
            font-family: Arial, sans-serif;
        }

        .card:hover {
            background: #f5f5f5;
            box-shadow: 5px 5px 12px #888;
        }

        .card button {
            background: #444;
            color: white;
            border: 1px solid #666;
            border-radius: 5px;
            padding: 6px 22px;
            margin-top: 8px;
            box-shadow: 2px 2px 5px #aaa;
        }

        .card button:hover {
            background: #666;
            color: white;
            box-shadow: 3px 3px 8px #888;
        }
    </style>
</head>

<body>

    <header>
        <h1>My Styled Webpage</h1>
        <p>CSS3 Styling with HTML5 Canvas</p>
    </header>

    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Services</a>
        <a href="#">Contact</a>
    </nav>

    <hr>

    <h2>Canvas Graphics</h2>

    <canvas id="myCanvas" width="450" height="200"></canvas>

    <script>
        var canvas = document.getElementById("myCanvas");
        var ctx = canvas.getContext("2d");

        ctx.fillStyle = "blue";
        ctx.fillRect(30, 30, 100, 60);
        ctx.fillStyle = "black";
        ctx.font = "12px Arial";
        ctx.fillText("Rectangle", 45, 110);

        ctx.beginPath();
        ctx.fillStyle = "red";
        ctx.arc(210, 60, 40, 0, 2 * Math.PI);
        ctx.fill();
```

```
        ctx.fillStyle = "black";
        ctx.fillText("Circle", 195, 120);

        ctx.beginPath();
        ctx.strokeStyle = "green";
        ctx.lineWidth = 4;
        ctx.moveTo(320, 30);
        ctx.lineTo(400, 90);
        ctx.stroke();
        ctx.fillStyle = "black";
        ctx.fillText("Line", 345, 120);

        ctx.fillStyle = "black";
        ctx.font = "bold 22px Arial";
        ctx.fillText("Harsha Vardhan", 150, 170);
    </script>

    <hr>

    <div class="container">
        <div class="card">
            <h3>Web Dev</h3>
            <p>HTML, CSS, JavaScript</p>
            <button>Learn</button>
        </div>

        <div class="card">
            <h3>Design</h3>
            <p>Graphics & UI/UX</p>
            <button>Learn</button>
        </div>

        <div class="card">
            <h3>Marketing</h3>
            <p>SEO & Social Media</p>
            <button>Learn</button>
        </div>
    </div>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
`linear-gradient(white, #e8e8e8)`	Creates a smooth vertical background color transition.
`display: flex`	Formats child .card elements in a flexible horizontal row layout.
`box-shadow: 3px 3px 8px #aaa`	Adds a soft drop shadow effect behind card components.
`canvas.getContext('2d')`	Retrieves the 2D rendering context interface for drawing on canvas.
`fillRect(x, y, w, h)`	Draws a filled rectangle at specified coordinates and dimensions.
`arc(x, y, r, sAngle, eAngle)`	Defines a circular path specified by center, radius, and angles.
`moveTo(), lineTo(), stroke()`	Creates a straight line path and renders it with specified stroke width/color.
`fillText(text, x, y)`	Draws text strings at specific (x, y) coordinates on the canvas.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load & Gradient	Load page in browser	Linear gradient background & header render cleanly	**Pass**
TC-02	Nav Bar Hover	Hover over nav links	Background changes to #ddd & border darkens	**Pass**

TC-03	Canvas Drawing	Inspect canvas element	Blue rectangle, red circle, & green line render	**Pass**
TC-04	Canvas Text	Inspect canvas text	Shape labels & bold name text display correctly	**Pass**
TC-05	Flexbox Cards Layout	Check card alignment	Cards display side-by-side with shadows & hover	**Pass**

9. OUTPUT

Output 1: Styled Webpage with HTML5 Canvas Graphics & CSS3 Flexbox Cards

10. RESULT

Thus, the web page demonstrating CSS3 styling, Flexbox layout, hover effects, and HTML5 Canvas 2D graphics rendering was successfully designed, implemented, and verified.